

The five-digit irrational-mask example is never spectral

Candidate full negative answer to Example 5.2 of arXiv:2209.05619

11 August 2026

Status. Candidate full proof, likely valid, pending independent expert review. In fact, the proof rules out even an infinite orthogonal family of exponentials for every contraction ratio.

1 Source problem

An, He, and Lai end *Classification of spectral self-similar measures with four-digit elements* (arXiv:2209.05619) with the first genuinely irrational five-digit test case [1, Example 5.2]. For

$$\mathcal{D} = \{0, 1, 3, 5, 6\}, \quad 0 < \rho < 1,$$

let $\mu_{\rho, \mathcal{D}}$ be the equal-weight self-similar measure for the maps $x \mapsto \rho(x + d)$, $d \in \mathcal{D}$. The source records that all real zeros of the mask are irrational and states:

“We believe that none of such $\mu_{\rho, \mathcal{D}}$ can be a spectral measure, but a more intensive study will be needed in this case.”

The (modified) Laba-Wang conjecture becomes a difficult problem to study if the number of digits is more than four because some irrationality phenomenon began to exist when the number of digits is five or more. The following example provides a case with purely irrational roots with five digits. It can be found in [1].

Example 5.2. (See [1, Example 7, p.1207]) Let $\mathcal{D} = \{0, 1, 3, 5, 6\}$. Then $M_{\mathcal{D}}(\xi)$ consists only of irrational roots for ξ . By periodicity,

$$\mathcal{Z}(M_{\mathcal{D}}) = \Theta + \mathbb{Z}$$

where $\Theta \subset \mathbb{R} \setminus \mathbb{Q}$. In this case, an easy argument shows that there does not exist any $\alpha > 0$ such that $\mathcal{Z}(M_{\mathcal{D}}) \subset \alpha\mathbb{Z}$. Therefore, the assumption for Theorem 1.3 is not satisfied. Hence, we cannot conclude immediately that $\mu_{\rho, \mathcal{D}}$ is spectral measure implies that ρ^{-1} is an integer. We believe that none of such $\mu_{\rho, \mathcal{D}}$ can be a spectral measure, but a more intensive study will be needed in this case.

We prove this belief for every $0 < \rho < 1$. The argument also gives a general obstruction whenever an integral mask has no rational zero phase.

2 Proof intuition

The Fourier transform is an infinite mask product. Hence every difference of two orthogonal frequencies has the form

$$q^k(\theta + n), \quad q = \rho^{-1} > 1, \text{quad}k \geq 1, \text{quad}n \in \mathbb{Z},$$

where θ is one of finitely many mask-zero phases. Infinite Ramsey theory gives an infinite subfamily for which the same θ works for every pair.

The phases in the source example are not merely irrational: their exponentials are algebraic, so Gelfond–Schneider makes the phases transcendental. If q is algebraic, coefficient comparison then forces pairwise differences of infinitely many powers of q to be powers of q , which is impossible by lacunarity.

If q is transcendental, one difference relation writes $\theta = R(q)$ for a nonconstant rational function $R \in \mathbb{Q}(X)$. Every other numerical difference identity then lifts, by transcendence of q , to an identity in $\mathbb{Q}(X)$. A degree-at-infinity lemma shows that the set $\{X^k(R+n)\}$ cannot contain an infinite one-sided difference clique. This eliminates the remaining case.

3 Preliminaries

For a finite integral digit set $\mathcal{D}$, write

$$M_{\mathcal{D}}(\xi) = \frac{1}{\#\mathcal{D}} \sum_{d \in \mathcal{D}} e^{-2\pi i d \xi}, \quad \Theta = \mathcal{Z}(M_{\mathcal{D}}) \cap [0, 1).$$

The Fourier transform of the associated equal-weight self-similar measure is

$$\widehat{\mu}_{\rho, \mathcal{D}}(\xi) = \prod_{j=1}^{\infty} M_{\mathcal{D}}(\rho^j \xi). \quad (1)$$

For fixed ξ , the product is nonzero if none of its factors is zero, because $M_{\mathcal{D}}(t) = 1 + O(t)$ at the origin and $\sum_j |\rho^j \xi| < \infty$. Therefore, with $q = \rho^{-1}$,

$$\mathcal{Z}(\widehat{\mu}_{\rho, \mathcal{D}}) = \bigcup_{\theta \in \Theta} \bigcup_{k \geq 1} q^k(\theta + \mathbb{Z}). \quad (2)$$

If $E(\Lambda) = \{e^{2\pi i \lambda x} : \lambda \in \Lambda\}$ is orthogonal in $L^2(\mu)$, then

$$(\Lambda - \Lambda) \setminus \{0\} \subseteq \mathcal{Z}(\widehat{\mu}). \quad (3)$$

Lemma 1 (Transcendence of irrational integral-mask phases). *Let $\mathcal{D} \subset \mathbb{Z}$ be finite. If $\theta \in \mathcal{Z}(M_{\mathcal{D}})$ is irrational, then θ is transcendental.*

Proof. The number $z = e^{-2\pi i \theta}$ is a zero of the integral polynomial $\sum_{d \in \mathcal{D}} z^d$ (after multiplying by a power of z if necessary), so z is algebraic. If θ were algebraic irrational, then Gelfond–Schneider, applied to the algebraic base -1 and algebraic irrational exponent -2θ , would say that every value of $(-1)^{-2\theta}$ is transcendental. Taking $\log(-1) = \pi i$ gives the value $e^{-2\pi i \theta} = z$, a contradiction. $\square$

The next lemma is the formal obstruction needed only when q is transcendental. For a nonzero rational function F , let $\deg_{\infty} F$ be numerator degree minus denominator degree in reduced form.

Lemma 2 (Rational-function difference-clique obstruction). *Let $R \in \mathbb{Q}(X) \setminus \mathbb{Q}$ and*

$$\mathcal{S}_R = \{X^k(R+n) : k \geq 1, n \in \mathbb{Z}\} \subset \mathbb{Q}(X).$$

There is no infinite sequence of distinct elements $f_0, f_1, \dots \in \mathcal{S}_R$ such that

$$f_j - f_i \in \mathcal{S}_R \quad (0 \leq i < j). \quad (4)$$

Proof. We use only the behavior at $X = \infty$.

First suppose R has a pole there, of order $d > 0$. Every $X^k(R + n)$ has degree $k + d$. An infinite fixed- k subfamily is impossible: the difference of two members is $(n' - n)X^k$, while equality with $X^c(R + \ell)$ forces $c = k - d$ and $R + \ell = (n' - n)X^d$; three distinct n 's would force two different coefficients for the same R . Thus, after passing to a subsequence, its levels k_j tend to infinity. Fix f_0 . For all large j the degree of $f_j - f_0$ is $k_j + d$, so (4) has level k_j and

$$X^{k_j}(R + n_j) - f_0 = X^{k_j}(R + \ell_j).$$

Hence $f_0 = (n_j - \ell_j)X^{k_j}$, which cannot hold for two distinct unbounded k_j .

Now suppose $L = R(\infty) \in \mathbb{Q}$. Put $H = R - L$. A level (k, n) is *normal* if $L + n \neq 0$; it then has degree k . If $-L \in \mathbb{Z}$, there is also one exceptional choice $n_* = -L$, for which $X^k(R + n_*) = X^k H$. Write $\deg_\infty H = -d < 0$.

Assume first that infinitely many f_j are normal. As in the preceding paragraph, an infinite fixed-level subfamily is impossible (differences are integer multiples of a monomial, while a possible exceptional representation fixes that multiple). Pass to normal levels $k_j \rightarrow \infty$ and fix f_0 . If $f_j - f_0 = X^{c_j}(R + \ell_j)$ is normal, degrees give $c_j = k_j$ and then $f_0 = (n_j - \ell_j)X^{k_j}$, again impossible twice. If it is exceptional, then $c_j = k_j + d$ and

$$f_0 = X^{k_j}[(L + n_j) + H - X^d H]. \quad (5)$$

As n varies, the bracketed rational functions differ only by integer constants. All but at most one have degree zero; the remaining one has a single fixed finite degree. Their degrees therefore cannot equal $\deg_\infty f_0 - k_j \rightarrow -\infty$, contradicting (5).

It remains to consider infinitely many exceptional terms $X^{k_j}H$; their levels are necessarily distinct and unbounded. Fix k_0 . If

$$H(X^{k_j} - X^{k_0}) = X^c(R + \ell)$$

has an exceptional right side, cancellation of H would give $X^{k_j} - X^{k_0} = X^c$, an impossible Laurent-polynomial identity. For a normal right side, degrees give $c = k_j - d$, and rearrangement yields

$$H(X^d - 1 - X^{k_0 - k_j + d}) = L + \ell \in \mathbb{Q}. \quad (6)$$

Three distinct values of k_j in (6), after subtracting pairs, would make the ratio of two distinct binomials

$$\frac{X^a - X^b}{X^a - X^c}$$

a rational constant, which it is not. Thus this case is finite as well. All possibilities contradict infinitude. $\square$

4 The general obstruction

Theorem 3. *Let $\mathcal{D} \subset \mathbb{Z}$ be finite with at least two elements, and suppose every phase in $\Theta = \mathcal{Z}(M_{\mathcal{D}}) \cap [0, 1)$ is irrational. Then, for every $0 < \rho < 1$, the space $L^2(\mu_{\rho, \mathcal{D}})$ contains no infinite orthogonal family of exponentials. In particular, $\mu_{\rho, \mathcal{D}}$ is not spectral.*

Proof. Suppose Λ is an infinite orthogonal set. Color each unordered pair of distinct elements of Λ by one phase $\theta \in \Theta$ occurring in a representation (2) of the absolute difference (choose one deterministically if there is more than one). The set of colors is finite. By the infinite Ramsey

theorem, there is an infinite monochromatic subset. It has an infinite strictly monotone subsequence; reverse its sign if needed and write it as

$$\gamma_0 < \gamma_1 < \gamma_2 < \cdots.$$

For one fixed $\theta \in \Theta$, every positive difference is therefore

$$\gamma_j - \gamma_i = q^{k_{ij}}(\theta + n_{ij}), \quad i < j, \text{quad} k_{ij} \geq 1, \text{quad} n_{ij} \in \mathbb{Z}. \quad (7)$$

By Lemma 1, θ is transcendental.

Translate by γ_0 and set

$$d_i = \gamma_i - \gamma_0 = q^{k_i}(\theta + n_i) \quad (i \geq 1).$$

If q is algebraic, compare $d_j - d_i$ with (7). Since θ is transcendental over the algebraic numbers, its coefficient must vanish:

$$q^{k_j} - q^{k_i} = q^{k_{ij}} \quad (i < j). \quad (8)$$

Thus $k_j > k_i$. Fix $i = 1$ and let $j \rightarrow \infty$. Dividing (8) by q^{k_j} gives

$$1 - q^{k_1 - k_j} = q^{k_{1j} - k_j}.$$

The right side is a power of q strictly below 1, hence at most q^{-1} , whereas the left side tends to 1 from below. This is impossible.

It remains that q is transcendental. Use $d_2 - d_1$ once. On collecting the coefficient of θ , we obtain

$$(q^{k_2} - q^{k_1} - q^{k_{12}})\theta = B(q)$$

for a Laurent polynomial $B \in \mathbb{Z}[X, X^{-1}]$. The coefficient polynomial is not identically zero (its value at $X = 1$ is -1), and it cannot vanish at the transcendental number q . Hence

$$\theta = R(q)$$

for some $R \in \mathbb{Q}(X)$. This R is nonconstant because θ is irrational.

Replace q formally by X and define

$$f_i = X^{k_i}(R + n_i) \in \mathcal{S}_R.$$

Every equality (7), after substituting $\theta = R(q)$, is an equality of rational functions evaluated at the transcendental number q . A nonzero rational function over $\mathbb{Q}$ cannot vanish at q , so each equality lifts identically to $\mathbb{Q}(X)$. Consequently $f_j - f_i \in \mathcal{S}_R$ for all $i < j$, contradicting Lemma 2.

Thus no infinite orthogonal exponential set exists. The self-similar attractor is infinite when $\#\mathcal{D} \geq 2$, so $L^2(\mu)$ is infinite-dimensional; a finite orthogonal family cannot be an orthonormal basis. Hence the measure is not spectral. $\square$

Corollary 4 (The source's five-digit example). *For $\mathcal{D} = \{0, 1, 3, 5, 6\}$ and every $0 < \rho < 1$, the measure $\mu_{\rho, \mathcal{D}}$ is not spectral. More strongly, it admits no infinite orthogonal set of exponentials.*

Proof. The source paper records $\mathcal{Z}(M_{\mathcal{D}}) = \Theta + \mathbb{Z}$ with $\Theta \subset \mathbb{R} \setminus \mathbb{Q}$. Apply Theorem 3. $\square$

5 Verification, scope, and novelty

Proof stress tests. The Fourier-product zero equality uses absolute convergence near the origin, so it introduces no spurious zeros. Ramsey colors only by the finite phase, not by the unbounded scale. The algebraic- q argument covers all algebraic contractions, not merely reciprocal integers. In the transcendental case, evaluation at q is injective on $\mathbb{Q}(X)$; the rational-function lemma treats separately poles at infinity, ordinary finite limits, and the sole possible exceptional integer translate. Zero mask phases and repeated representations cause no issue because the source phases are irrational and the coloring chooses one representation.

Scope. Corollary 4 completely proves the explicit belief in Example 5.2, for all contraction ratios. Theorem 3 also covers every finite integral digit set whose mask has no rational zero phase. It does not settle the full modified Łaba–Wang conjecture when rational and irrational mask phases coexist.

Bounded novelty check. On 11 August 2026 we searched the run’s cheap registry/problem indexes, the exact digit set, the exact source sentence, the source title and citations, and close phrases involving irrational mask zeros and infinite orthogonal exponentials. We found the source paper and later works citing its four-digit classification, but no later resolution of Example 5.2 or statement matching Theorem 3. The ingredients (infinite Ramsey and Gelfond–Schneider) are classical; their combination with the rational-function difference-clique lemma appears new. Novelty confidence is moderate, while proof confidence is high.

Human-review recommendation. Verify the degree-at-infinity case split in Lemma 2, especially the exceptional translate $n = -R(\infty)$; then check the lifting of numerical identities to $\mathbb{Q}(X)$ when q is transcendental. The included symbolic script is only a sanity check and is not used as proof.

References

- [1] L.-X. An, X. He, and C.-K. Lai, *Classification of spectral self-similar measures with four-digit elements*, Asian J. Math. 27 (2023), 467–492; arXiv:2209.05619.
- [2] A. Baker, *Transcendental Number Theory*, Cambridge University Press, 1975.
- [3] F. P. Ramsey, *On a problem of formal logic*, Proc. London Math. Soc. (2) 30 (1930), 264–286.